1. (Fall 2024 #6) Consider the function: $f(x) = 2x^{5/3} + 5x^{2/3}$.

 (a) Find the x-values of all local maxima and local minima of $f(x)$ on its domain.

$$f'(x) = 2 \cdot \frac{5}{3} x^{\frac{2}{3}} + 5 \cdot \frac{2}{3} x^{-\frac{1}{3}} = \frac{10}{3} x^{-\frac{1}{3}} (x+1)$$

$f'(x) = 0$	$f'(x)$ DNE
$x = -1$	$x = 0$

Number line: -2, -1, $-\frac{1}{2}$, 0, 1

	$x<-1$	$-1<x<0$	$x>0$
$f'(x)$	$+$	$-$	$+$
$f(x)$	$\nearrow$	$\searrow$	$\nearrow$

test points to check the sign of f'

$$f'(-2) = \frac{10}{3} \underbrace{(-2)^{-\frac{1}{3}}}_{-} \underbrace{(-2+1)}_{-} = +$$

$$f'(-\tfrac{1}{2}) = \frac{10}{3} \underbrace{(-\tfrac{1}{2})^{-\frac{1}{3}}}_{-} \underbrace{(-\tfrac{1}{2}+1)}_{+} = -$$

$$f'(1) = \frac{10}{3} \cdot \underbrace{1^{-\frac{1}{3}}}_{+} \underbrace{(1+1)}_{+} = +$$

local max: at $x = -1$

local min: at $x = 0$

 (b) Find the absolute maximum value of $f(x)$ on the interval $[-1, 1]$.

Recall: (Extreme Vaue Theorem)

$f(x)$ is continuous on the closed interval $[-1,1]$,

$\Downarrow$

f has absolute extrema, and these extrema must occur either at the endpoints or at critical points.

Therefore, we only need to evaluate f at the endpoints and critical points.

$$\left.\begin{aligned} f(-1) &= 2 \cdot (-1) + 5 \cdot 1 = 3 \\ f(0) &= 0 \\ f(1) &= 2 \cdot 1 + 5 \cdot 1 = 7 \end{aligned}\right]$$

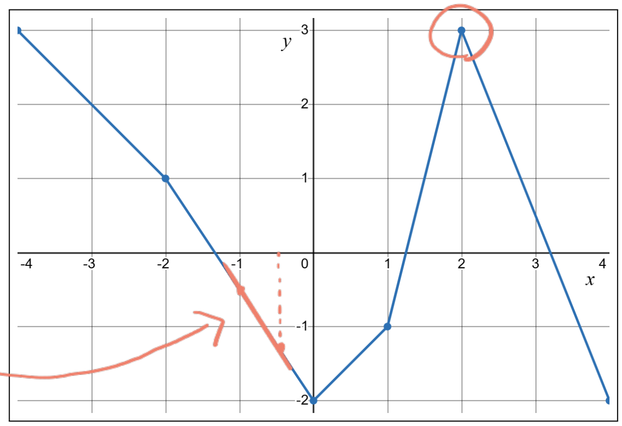
$0 < 3 < 7 \Rightarrow$ abs min: at $x = 0$, $f(0) = 0$

abs max: at $x = 1$, $f(1) = 7$

2. (Fall 2020, #7) Consider the following graph:

sharp corner $\Rightarrow f$ is not differentiable at $x=2$. $f'(2)$ DNE.

the slopes of the tangent lines at $x=-1$ and $x=-\frac{1}{2}$ are both $-\frac{3}{2}$



(a) Suppose that this is the graph of a function $f = f(x)$. In particular, $f(-1) = -1/2$.

i. Consider the function $g(x) = f(3x - 4)$. Compute $g'(2)$, if possible. If this is not possible, then explain why.

$$g'(x) \underset{\uparrow \text{ chain rule}}{=} 3f'(3x-4)$$

$$g'(2) = 3\underbrace{f'(2)}_{\text{DNE}} = \text{DNE}$$

ii. Consider the function $h(x) = f(f(x)) + \ln(1 - \tan(\pi f(x)/2))$. Compute $h'(-1)$, if possible. If this is not possible, then explain why.

$$h'(x) = f'(f(x))f'(x) + \frac{1}{1-\tan\left(\frac{\pi f(x)}{2}\right)} \cdot \left(-\sec^2\left(\frac{\pi f(x)}{2}\right) \cdot \frac{\pi f'(x)}{2}\right)$$

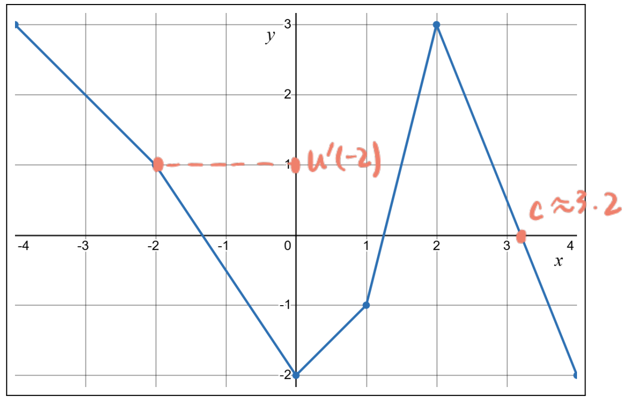
evaluate at $x=-1$: (note that we are given $f(-1) = -\frac{1}{2}$)

$$h'(-1) = \underbrace{f'\left(-\tfrac{1}{2}\right)}_{-\frac{3}{2}}\underbrace{f'(-1)}_{-\frac{3}{2}} + \frac{1}{1-\tan\left(-\frac{\pi}{4}\right)}\left(\underbrace{-\sec^2\left(-\tfrac{\pi}{4}\right)}_{= \frac{1}{\cos^2\left(-\frac{\pi}{4}\right)}} \cdot \frac{\pi f'(-1)}{2}\right)$$

$$= \frac{9}{4} + \frac{1}{1-(-1)}\left(-\frac{1}{\left(\frac{\sqrt{2}}{2}\right)^2} \cdot \frac{\pi}{2} \cdot \left(-\frac{3}{2}\right)\right)$$

$$= \frac{9}{4} + \frac{1}{2} \cdot 2 \cdot \frac{3\pi}{4}$$

$$= \boxed{\frac{9}{4} + \frac{3\pi}{4}}$$



(b) Now suppose that this is the graph of the **derivative** u' of a function $u = u(x)$.

i. Compute an approximate value of $u(-2.1)$. Show your work.

Recall the linear approximation of $u(x)$ at $x = a$: $L(x) = u(a) + u'(a)(x-a)$

To approximate $u(-2.1)$, we choose $a = -2$ since it is close to -2.1, and the value $u'(-2)$ is easily read from the given graph, $u'(-2) = 1$.

Suppose $u(-2) = C$ for some constant C, then

$$L(-2.1) = C + 1\cdot(-2.1+2) = C - 0.1$$

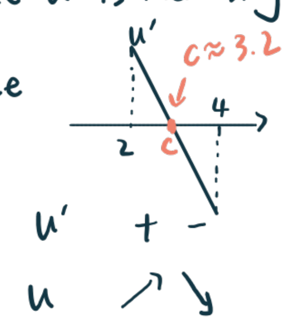
ii. For each of the following statements, indicate whether it is true, false, or you need more information to make the conclusion, and give a brief explanation.

① u is concave up on $(0, 2)$ **True** **False** **Need More Info**

② u has a local maximum on $[2, 4]$ **True** **False** **Need More Info**

① is true since u' is increasing on $(0, 2)$

② is true since



u' is positive on $(2, c)$, negative on $(c, 4)$] indicating a critical point at $x = c$ where u changes from increasing to decreasing.

3. (Fall 2020, #10) A clothing store produces and sells suits. The total (cumulative) cost, in dollars, of producing q suits is

$$C(q) = 4000 + 0.25q^2$$

To sell q suits, the price per suit, in dollars, must be

$$p = 150 - 0.5q$$

(a) What is the profit from selling q suits?

$$\begin{aligned} f(q) = \text{profit} &= (\#\text{ suits sold}) \times \text{unit price} - \text{total cost} \\ &= qp - C(q) \\ &= q(150 - 0.5q) - (4000 + 0.25q^2) \\ &= -4000 + 150q - 0.75q^2 \end{aligned}$$

(b) How many suits must the store produce and sell to maximize profit?

$$f'(q) = 150 - 1.5q$$

$f'(q) = 0$	$f'(q)$ DNE
$q = 100$	none

		100	
f'	$+$		$-$
f	$\nearrow$		$\searrow$

$q = 100$ is the only local max, and also the abs max

(c) What is this maximum profit?

$$f(100) = -4000 + 15000 - 7500 = 3500 \text{ (dollars)}$$

(d) What price per suit must the store charge to maximize profit?

$$p(100) = 150 - 0.5 \times 100 = 100 \text{ (dollars)}$$

4. (Fall 2019, #1) Consider the function $f(x) = 3x^{\frac{5}{3}} + 60x^{\frac{2}{3}}$.

(a) Show that $f'(x) = \dfrac{5(x+8)}{\sqrt[3]{x}}$. Then find and classify all local extrema of f.

$$f'(x) = 3 \cdot \frac{5}{3} x^{\frac{2}{3}} + 60 \cdot \frac{2}{3} x^{-\frac{1}{3}} = 5x^{\frac{2}{3}} + 40x^{-\frac{1}{3}} = 5x^{-\frac{1}{3}}(x+8) = \frac{5(x+8)}{\sqrt[3]{x}}$$

$f'(x) = 0$	$f'(x)$ DNE
$x = -8$	$x = 0$

	-10	-8	-1	0	1
$f'(x)$	$+$		$-$		$+$
$f(x)$	$\nearrow$		$\searrow$		$\nearrow$

test points
$$f'(-10) = \frac{5(-10+8)}{\sqrt[3]{-10}} = \frac{(-)}{(-)} = +$$
$$f'(-1) = \frac{(+)}{(-)} = -$$
$$f'(1) = \frac{(+)}{(+)} = +$$

local max: at $x = -8$

local min: at $x = 0$

Extreme Value Theorem

(b) Find the values of the global extrema of f on the domain $[-1, 1]$.

Since $f(x)$ is continuous everywhere and $[-1, 1]$ is a closed interval, the EVT applies.
We only need to check the function values at the endpoints and critical points:

$$f(-1) = 3(-1)^{\frac{5}{3}} + 60(-1)^{\frac{2}{3}} = 57$$
$$f(0) = 0$$
$$f(1) = 3 + 60 = 63$$

$0 < 57 < 63$

$$(-1)^{\frac{5}{3}} = \left((-1)^5\right)^{\frac{1}{3}} = (-1)^{\frac{1}{3}} = -1$$
$$(-1)^{\frac{2}{3}} = \left((-1)^2\right)^{\frac{1}{3}} = (1)^{\frac{1}{3}} = 1$$

So: abs min: $f(0) = 0$
abs max: $f(1) = 63$

(c) Justify briefly, but precisely, why f has global extrema on the interval $[-1.1]$.

Since $f(x)$ is continuous everywhere and $[-1, 1]$ is a closed interval, the EVT applies.

So f attains global extrema on $[-1, 1]$.

5. (Fall 2017, #7) Find all points in the interval $[-2,2]$ on which the function

$$f(x) = \frac{x^{\frac{2}{3}}}{2+x^2}$$

has a local maximum or minimum, and find the absolute maximum and minimum values of f on this interval.

$$f'(x) = \frac{\frac{2}{3}x^{-\frac{1}{3}}(2+x^2) - 2x\cdot x^{\frac{2}{3}}}{(2+x)^2}$$

← we only need to analyze the numerator, whether $f'=0$ or DNE is entirely determined by the numerator.

← denominator is always >0, it does not affect the sign of f'

$$\text{numerator} = \frac{4}{3}x^{-\frac{1}{3}} + \frac{2}{3}x^{\frac{5}{3}} - 2x^{\frac{5}{3}} = \frac{4}{3}x^{-\frac{1}{3}} - \frac{4}{3}x^{\frac{5}{3}} = \frac{4}{3}x^{-\frac{1}{3}}(1-x^2) = \frac{4}{3}x^{-\frac{1}{3}}(1+x)(1-x) = \frac{4}{3}\,\frac{(1+x)(1-x)}{x^{\frac{1}{3}}}$$

$f'(x)=0$	$f'(x)$ DNE
$x=\pm 1$	$x=0$

test points: -1.5, -0.5, 0.5, 1.5

x	-2		-1		0		1		2
$f'(x)$		$+$		$-$		$+$		$-$	
$f(x)$		↗		↘		↗		↘	

test points

$$f'(-1.5) = \frac{4}{3}\,\frac{\overbrace{(1+(-1.5))}^{(-)}\overbrace{(1-(-1.5))}^{(+)}}{\underbrace{(-1.5)^{\frac{1}{3}}}_{(-)}} = +$$

$$f'(-0.5) = \frac{(+)(+)}{(-)} = -$$

$$f(0.5) = \frac{(+)(+)}{(+)} = +$$

$$f(1.5) = \frac{(+)(-)}{(+)} = -$$

So: local min : at $x=-2, 0, 2$

local max : at $x=-1, 1$

To find the absolute extrema: since f is continuous on the closed interval $[-2,2]$, we only need to check the endpoints and critical points:

$$f(-2) = \frac{(-2)^{\frac{2}{3}}}{6} = \frac{2^{\frac{2}{3}}}{6} < \frac{2^1}{6} = \frac{1}{3}$$

↑ since $\frac{2}{3}<1$

$$f(-1) = \frac{(-1)^{\frac{2}{3}}}{3} = \frac{1}{3}$$

$$f(0) = 0$$

$$f(1) = \frac{1}{3}$$

$$f(2) = \frac{2^{\frac{2}{3}}}{6} < \frac{2^1}{6} = \frac{1}{3}$$

$$0 < \frac{2^{\frac{2}{3}}}{6} < \frac{1}{3}$$

So: abs max: $f(-1) = f(1) = \frac{1}{3}$

abs min: $f(0) = 0$

$$(-2)^{\frac{2}{3}} = \left((-2)^2\right)^{\frac{1}{3}} = (2^2)^{\frac{1}{3}} = 2^{\frac{2}{3}}$$

$$(-1)^{\frac{2}{3}} = \left((-1)^2\right)^{\frac{1}{3}} = 1^{\frac{1}{3}} = 1$$